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USING ON-BODY MICROPHONE TO IMPROVE USER INTERACTION WITH SMART DEVICES

Abstract

Methods, devices, and processor-readable media for adjusting one or both of a listening and talking experience, including processing a user device audio signal representing a sound segment received at a microphone of an on-body user device of a user, wherein the sound segment is propagated as an acoustic wave between a smart device and the user and performing an action based on the processing.

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Background/Summary

RELATED APPLICATION DATA [0001] The present disclosure is a continuation of PCT Application No. PCT/CN2022/143796, filed on Dec. 30, 2022, entitled “USING ON-BODY MICROPHONE TO IMPROVE USER INTERACTION WITH SMART DEVICES”, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present application generally relates methods, systems and computer media related to using an on-body microphone to improve user interaction with smart devices.

BACKGROUND

[0003] The concept of a “Smart Home” that includes several computer-enabled smart devices connected to a wireless home network has been popular for several years. Examples of common smart devices that can be included in a smart home environment include smart TVs, virtual assistant-enabled smart speakers, smart lights, smart electrical switches and smart appliances such as smart fridges, which support audio notifications and voice commands.

[0004] Core experiences for users in a Smart Home environment include “Listen” and “Talk” experiences. As used here, the “Listen” experience can refer to the consumption and enjoyment of a user of the audio content provided by smart home devices; the “Talk” experience can refer to voice inputs provided by a user to smart home devices, which for example can include audio commands to such devices or the use of such devices as a communication intermediary between the user and other users, including their family or friends. Such experiences are quite important for the Smart Home environment because Smart Home devices are typically equipped with built-in microphones and speakers, which allow voice-based interaction.

[0005] Existing smart home devices have a number of operational shortfalls in the context of “Listen” and “Talk” experiences. In the case of the “Listen” experience, output audio settings (including volume and sound effect settings) are typically pre-defined at pre-set levels for each smart device such that a user must manually adjust the settings to make changes. Furthermore, even if a user wants to adjust these audio settings (e.g., through a control panel on the smart device, through a connected smartphone interface, or through a smart device remote control unit), they can only set the audio settings for the current audio environment that they are in, which is a factor of the specific region the user is physically located in and the current background noise conditions of that location. Once the current audio environment changes (due to one or both of user movement from a region or a change in background noise), the user selected audio settings may be imperfect for the user's new audio environment. Moreover, as there are normally multiple users in a smart home environment, and each user may have their own preference for the sound settings. Therefore when user starts using a smart device (e.g., a smart TV or smart speaker) after another user, they may find the sound settings are far from their preference.

[0006] In the case of the “Talk” experience, smart devices need to receive and detect users' voices to support voice interaction, and the threshold for voice reception (or the gain of the smart device microphone) is also pre-set or defined by the manufacturers. The location of a user and ambient noise can both impact the “Talk” experience, requiring a user to change their tone and voice in

order for the smart device to recognize and understand their voice.

[0007] Solutions have been proposed to improve the smart home experience. Such solutions can, for example, be configured to recognize the location, movement direction, and identity of users present in a smart home environment. Two of these solutions include (i) using camera and computer vision techniques, and (ii) using laser and radar techniques.

[0008] Regarding solutions that are based on camera and computer vision techniques, such solution rely on smart devices equipped with built-in cameras for collecting visual light images and/or infrared light images. Through processing and analyzing these images, information about a surrounding environment the device can be deduced, including the presence of objects and/or users around the device and the distances between them. This image based information can be used for adjusting sound-based interaction. For example, the CMU-BC1 BRAVIA CAM™ by Sony™ has a feature called Ambient Optimization Pro that controls aspects of a smart TV sound output based on where a user is a room and how far the user is from the TV. Based on user position, left and right sound bounce will be automatically.

[0009] However, image based solutions can raise privacy concerns for some users. To achieve the high accuracy of detection, the system has to have an always-on camera, and an always-on camera can over-collect information, making users feel uncomfortable. This discomfort can be enhanced by fears of a possible hacking of the camera. As a result, many users prefer to keep the lens of smart device cameras covered unless they specifically require the camera such as for a video conference session. In addition to security issues, many smart device are not equipped with cameras and thus camera based solutions are not ubiquitous. Further, cameras must be aimed to capture a relevant field-of-view direction that includes the user in order to provide user-location-based functionality.

[0010] In the case of Lidar or radar based solutions to detect user distance and location, smart devices such as smart cleaning robots are able to construct a map of their environment and then perform real-time comparisons to detect obstacles. Although Lidar and radar sensors are capable of producing accurate results, they are expensive and can be relatively large. In addition, the energy consumption of Lidar or radar sensors is relatively high. For devices such as robots with specific usage purposes, the tradeoff between accuracy, cost and energy consumption can be acceptable. However, Lidar or radar based solutions may be economically or technically impractical for many types of smart devices.

[0011] Accordingly, there is a need for methods and systems that can provide an effective and efficient user-specific “Listen” and “Talk” experiences when it comes to sound in a smart-home environment.

SUMMARY

[0012] The present disclosure describes methods and devices for providing one or both of enhanced or both of a listen and a talk experience in a smart environment by capturing audio data of a sound using a microphone of an on-body user device.

[0013] According to example aspects an automated method for enhancing one or both of a listen and a talk experience is disclosed. The method includes: processing a user device audio signal representing a sound segment received at a microphone of an on-body user device of a user, wherein the sound segment is propagated as an acoustic wave between a smart device and the user; and performing an action based on the processing.

[0014] In some example aspects, the sound segment is generated by one or more speakers of the smart device and propagates from the one or more speakers to the user.

[0015] In some examples of one more of the preceding aspects, processing the user device audio signal comprises extracting information about one or more sound properties of the sound segment from the user device audio signal and performing the action comprises causing an adjustment of the one or more sound properties for future sounds generated by the one or more speakers.

[0016] In some examples of one more of the preceding aspects, the one or more sound properties

includes a volume.

[0017] In some examples of one more of the preceding aspects, the one or more sound properties includes a frequency equalization.

[0018] In some examples of one more of the preceding aspects, the one or more sound properties includes a relative balancing of multiple sound channels.

[0019] In some examples of one more of the preceding aspects, the one or more sound properties includes a relative time delay between multiple sound channels.

[0020] In some examples of one more of the preceding aspects, extracting information about one or more sound properties of the sound segment comprises comparing a reference audio signal used to generate the sound segment with the user device audio signal.

[0021] In some examples of one more of the preceding aspects, the method includes receiving by the smart device, through a wireless network, a transmitted copy of the user device audio signal, wherein processing the user device audio signal and performing the action are each performed by a processor system of the smart device.

[0022] In some examples of one more of the preceding aspects, the method includes sending by the smart device, through the wireless network, a message for the on-body user device requesting the on-body user device to record the sound segment to generate the user device audio signal.

[0023] In some examples of one more of the preceding aspects, the method includes generating, by the on-body user device, the user device audio signal, wherein processing the user device audio signal and performing the action are each performed by a processor system of the on-body user device and performing the action comprises causing an action request message to be sent through a wireless network communication link for the smart device.

[0024] In some examples of one more of the preceding aspects, the sound segment comprises a frequency sweep.

[0025] In some examples of one more of the preceding aspects, processing the user device audio signal comprises determining based on one or more parameters extracted from the user device audio signal when an unsafe listening experience exists for the user, and performing the action comprises, upon determining the unsafe listening experience exists, performing one or more actions to end the unsafe listening experience, including one or more of: causing a reduction in sound volume; adjusting a sound frequency equalization; causing sound to stop being generated; causing a message about the unsafe listening experience to be displayed on a display screen; and causing a message about the unsafe listening experience to be transmitted to a further device.

[0026] In some examples of one more of the preceding aspects, the method includes storing user device specific settings for the on-body user device and a plurality of further user devices, wherein performing the action is further based on the user device specific settings stored in respect of the on-body user device.

[0027] In some examples of one more of the preceding aspects, performing the action comprises causing one or both of a listening experience and a talking experience of the user to be automatically adjusted without requiring user action to effect the adjustment.

[0028] In some examples of one more of the preceding aspects, the sound segment originates from a mouth of the user and propagates from the user to a microphone of the smart device, and the method includes: generating, by the smart device, a local smart device audio signal representing the sound segment as received at the microphone of the smart device; receiving by the smart device, through a wireless network, a transmitted copy of the user device audio signal; and processing the local smart device audio signal in combination with processing the user device audio signal; wherein the action comprises generating a final audio signal based on the combined processing of the local smart device audio signal and the user device audio signal.

[0029] In some examples of one more of the preceding aspects, the smart device comprises one of a smart TV, an interactive smart speaker, a smart appliance, a smart sound system, and the on-body user device comprises one of a smart watch, a fitness tracker, smart glasses, a headset, an earphone,

smart clothes and a hand-held mobile device.

[0030] In some aspects, the present disclosure describes a non-transitory processor-readable medium having machine-executable instructions stored thereon which, when executed by a processor device, cause the processor device to perform any of the methods described above.

[0031] In some aspects, the present disclosure describes a non-transitory processor-readable medium having instructions tangibly stored thereon. The instructions, when executed by a processor device, cause the processor device to perform any of the methods described above.

[0032] The examples disclosed herein may provide various advantages. The disclosed solutions can, in some scenarios, provide users with optimized experience without users' manual operation of individual smart devices that are present in the environment. The disclosed examples provide a solution by using the microphone of an on-body user device to enhance the user experience. Such a solution does not require dedicated hardware such as radar or Lidar systems to operate, and can be less intrusive than a video capture based solution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] Reference will now be made, by way of example, to the accompanying drawings which show example embodiments of the present application, and in which:

[0034] FIG. 1 is a block diagram illustrating an example of a smart home environment to which example embodiments of methods and systems of the present disclosure can be applied;

[0035] FIG. 2 is a block diagram of an illustrative example of a smart device that can be used to implement one or more of the smart devices included in the smart home environment of FIG. 1;

[0036] FIG. 3 is a flow diagram showing operations performed by a first smart device and an on-body user device to adjust a user listening experience, according to an example embodiment;

[0037] FIG. 4 is a flow diagram showing operations performed by a first smart device and an on-body user device to adjust a user listening experience, according to a further example embodiment;

[0038] FIG. 5 is a flow diagram showing operations performed by a first smart device and an on-body user device to adjust a user listening experience, according to a further example embodiment; and

[0039] FIG. 6 is a flow diagram showing operations performed by a first smart device and an on-body user device to adjust a user talking experience, according to a further example embodiment.

[0040] Similar reference numerals may have been used in different figures to denote similar components.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0041] The present disclosure describes methods and devices for providing enhanced “Listen” and “Talk” experiences in a smart-home environment by recording sound using a microphone of on-body user device to provide a corresponding audio signal that can be processed to determine actions that can be performed to provide enhanced user experiences.

[0042] As used in the disclosure, sound is a vibration that propagates as an acoustic wave, through a transmission medium such as air. Human perceptible acoustic waves fall within the audio frequency range, between about 20 Hz and 20 kHz.

[0043] As used in this disclosure, an audio signal is a representation of sound. Examples of audio signals include analog audio signals and digital audio signals, which can for example be transmitted using electromagnetic signals or optical signals. A recorded or stored audio signal refers to an audio signal that is stored using a storage medium for future retrieval.

[0044] FIG. 1 is a block diagram illustrating an example of a smart home environment **100** to which example embodiments of methods and systems of the present disclosure can be applied. The environment **100** includes a plurality of smart devices. As used here, smart device can refer to a

processor-enabled device that can interact with a human user (e.g. user **102**) and interact with other electronic devices (e.g., other smart devices) via one or more wireless network protocols (e.g., Bluetooth™, Zigbee™, near-field communication, wireless local area network such as WiFi™ or LiFi™, and/or wide area networks such as LTE, 5G and 6G). The smart devices in smart home environment **100** can for example include one or more static smart devices that are generally in a fixed location when in use, including for example a smart TV **104** and an interactive smart speaker **106**. The environment **100** also includes one or more smart devices that are often mobile when in use, for example a smart vacuum (not shown) and wearable or on-body user devices such as smart glasses **108**, a smart earphone **110** (or headset), and a smart watch **112**. As used herein, on-body user device refers to a smart device that is intended to be commonly attached to, worn by, or carried by a human user **102**. A mobile smart phone **114** can function as an on-body device when it is held by a user **102** or fixed to the user **102** by an arm band or a belt holster or belt clip.

[0045] Other non-limiting examples of static smart devices that could be included in the environment **100** can include, among other things, smart lights, smart electrical switches, smart controllers, and smart appliances such as smart fridges. On-body user devices can include any type of smart device that a user wears or carries on their body. Other non-limiting examples of on-body user devices can include smart cloths, smart fitness trackers, and smart wearable fall detection devices, among other things.

[0046] In the illustrated example, smart TV **104** has a display screen **122** and a sound system **118** that includes one or more audio signal to sound transducers (e.g., speakers **124**, **126** and **128**). The sound system **118** can, for example be integrated into the housing of smart TV **104**. In alternative example, the sound system **118** can be a further smart device that includes one or more components housed separately from the smart TV **104**, but is coupled to receive audio signals from the TV to generate sounds for consumption by user **102**. In the illustrated example, sound system **118** may for example be a multi-channel system that generates multiple output directional sound channels using respective speakers. For example, speaker **124** can output sound corresponding to a left audio channel signal, speaker **126** can output sound corresponding to a center audio channel signal, speaker **128** can output sound corresponding to a right audio channel signal.

[0047] In the illustrated example, the sound system **118** includes a sound controller **150** that processes an input audio signal to provide respective audio channel signals for each of the sound channel specific speakers **124**, **126** and **128**. A number of properties of the sound generated by speakers **124**, **126** and **128** sound system **118** can be controlled by the sound controller **150**. For example, these sound properties can include one or more of: volume; equalization between frequency bands of the acoustic waves that collectively form the sound; balance and fade between the multiple sound channels; relative delay between the sound channels; and special effects applied to the sound.

[0048] In the illustrated examples, one or more of the smart devices included in the smart home environment **100** include built-in or associated microphones. For example, in the case of on-body user devices, smart glasses **108**, earphones **110**, smart watch **112** and smart phone **114** can each include an integrated microphone **116**. Static smart devices such as the smart TV **104** and smart speaker **106** can also include microphones **117**.

[0049] The microphones **116** associated with on-body user devices can be used to capture a representation of the sound experience of user **102** within the smart home environment **100**. In most scenarios, the closer that a microphone **116** is to the ears of the user **102**, the closer the sound captured by such a microphone will be to the user's actual sound experience. For example, an audio signal representation of a sound segment recorded by microphone **116** of earphone **110** or smart glasses **108** may, in some scenarios, be more representative of the user's sound experience than that of a wrist mounted or hand carried device.

[0050] The smart devices included in smart home environment **100** are able to communicate through a local or home wireless network **129** that can include one or more wireless networks that

use respective wireless protocols. For example, in some embodiments one or more of the on-body user devices may be enabled to exchange electronic signals with the smart TV **104** and smart speaker **106** using a wireless local area network (e.g., a Wi-Fi Network™) that includes a base station **120**. Base station **120** establishes respective communication links **130** with multiple respective smart devices and relays messages between these devices. In some examples, direct peer-to-peer wireless networks can be established using the Bluetooth™ protocol or Wi-Fi™ ad-hoc protocol, illustrated in FIG. **1** by communications links **132** and **134**, for example. In some examples, communication channels between smart devices may be established using a combination of network protocols. For example, earphone **110** may establish a Bluetooth™ peer-to-peer connection with smart device **114**, which in turn communicates with smart TV **104** via base station **120** using Wi-Fi communication links **130**.

[0051] As noted above, smart devices of smart home environment **100** are processor-enabled devices. FIG. **2** shows a block diagram of a representative processor system **200** that can be used to implement one or more of the respective smart devices (e.g., smart TV **104**, sound system **118**, smart speaker **106**, smart glasses **108**, earphones **110**, smart watch **112**, and smart phone **114**) of smart home environment **100**. Although an example embodiment of the processor system **200** is shown and discussed below, other embodiments may be used to implement examples disclosed herein, which may include components different from those shown. Although FIG. **2** shows a single instance of each component of the processor system **200**, there may be multiple instances of each component shown.

[0052] The processor system **200** includes one or more processors **202**, such as a central processing unit, a microprocessor, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a dedicated logic circuitry, a tensor processing unit, a neural processing unit, a dedicated artificial intelligence processing unit, or combinations thereof. The one or more processors **202** may collectively be referred to as a “processor device”. The processor system **200** also includes one or more input/output (I/O) interfaces **204**, which interfaces with input devices (e.g., camera **203**, microphone **116** or **117**) and output devices (e.g., display **122** and sound controller **150**). The processor system **200** may include other input devices (e.g., buttons, knobs, touchscreen, etc.) and other output devices (e.g., a diagnostic port enabling data input and output).

[0053] The processor system **200** includes one or more network interfaces **206** communication with the wireless network **129** of smart home environment **100**, including, for example, one or more interfaces for supporting wireless communication links **130**, **132** or **134**.

[0054] The processor system **200** includes one or more memories **208**, which may include a volatile or non-volatile memory (e.g., a flash memory, a random access memory (RAM), and/or a read-only memory (ROM)). The non-transitory memory(ies) **208** may store instructions for execution by the processor(s) **202**, such as to carry out examples described in the present disclosure. The memory(ies) **208** may include other software instructions, such as for implementing an operating system and other applications/functions. In the illustrated example, the memory **208** can include software instructions **160I** for implementing a smart device interaction module on the smart device. In some examples, the software instructions **160I** can be integrated into instructions for implementing a device operating system that supports interconnection between smart devices and the components of such devices, including, for example the Harmony Distributive OS™ available from Huawei™.

[0055] In some examples, the memory **208** may store associated smart device data **162** that includes the identities (e.g., device IDs) of other smart devices that are known to the smart device and preferred settings associated with those other devices.

[0056] In some examples, the processor system **200** may also include one or more electronic storage units (not shown), such as a solid state drive, a hard disk drive, a magnetic disk drive and/or an optical disk drive. In some examples, one or more data sets and/or modules may be provided by an external memory (e.g., an external drive in wired or wireless communication with the processor

system **200**) or may be provided by a transitory or non-transitory computer-readable medium. Examples of non-transitory computer readable media include a RAM, a ROM, an erasable programmable ROM (EPROM), an electrically erasable programmable ROM (EEPROM), a flash memory, a CD-ROM, or other portable memory storage. The components of the processor system **200** may communicate with each other via a bus, for example.

[0057] Referring again to FIG. **1**, the sound controller **150** of the sound system **118** of smart TV **104** can have one or more user interface controls that enable user control of output sound properties (e.g., volume; equalization between frequency bands of the acoustic waves that collectively form the sound; balance and fade between the multiple sound channels; relative delay between the sound channels; and special effects applied to the sound). Further, in the illustrated example, the sound system **118** includes a smart device interaction module **160**. As used here, a “module” can refer to a combination of a hardware processing circuit (e.g. processor **202**) and machine-readable instructions (software (e.g., smart device interaction instructions **160I**) and/or firmware) executable on the hardware processing circuit. A hardware processing circuit can include any or some combination of a microprocessor, a core of a multi-core microprocessor, a microcontroller, a programmable integrated circuit, a programmable gate array, a digital signal processor, or another hardware processing circuit.

[0058] The smart device interaction module **160** is configured to provide control signals to the sound controller **150** to cause the sound controller **150** to adjust one or more of the output sound properties. Example embodiments are described herein whereby the sound properties are dynamically and automatically adjusted based on the presence and location of the user **102** in the smart home environment **201**. In particular, a sound experience of the user **102** can be optimized by using the microphone **116** of one or more of the on-body user devices (e.g., smart glasses **108**, earphones **110**, smart watch **112** and/or smart phone **114**) associated with the user **102** to monitor the sound experience of the user **102** and provide feedback to the smart device interaction module **160**.

[0059] In this regard, FIG. **3** is a flow diagram showing operations **300** performed by a first smart device (first device operations **301**) and an on-body user device (target on-body user device operations **303**) to adjust a user listening experience, according to a first example embodiment. For illustrative purposes, operations **300** will be explained in the context of the sound system **118** of smart TV **104** functioning as the first smart device and smart glasses **108** functioning as the target on-body user device. Each of the sound system **118** and smart glasses **108** can have a respective smart device processor system such as smart device processor system **200** implementing a respective smart device interaction module **160** that configures the sound system **118** and smart glasses **108** to each respectively perform first device operations **301** and target on-body user device operations **303**.

[0060] In an example embodiment, a plurality of smart devices are known to (e.g., pre-registered with) the smart device interaction module of the **160** of the sound system **118**, and information for each of these devices is stored as associated smart device data. By way of example, Table 1 below is illustrative of the type of associated smart device data **162** that may be stored.

TABLE-US-00001

TABLE 1 Associated Smart Device Data		On- Sound Property		Settings		Device	
User	Connect	board	Channel	Channel ID	ID	Status	Mic
SG	108	102	YES	Y	EP	110	102
			YES	Y	SW	112	102
			YES	Y	SP	114	102
			YES	Y	...		

[0061] As shown in Table 1, the associated smart device data **162** can, for example include: a device ID for all pre-registered on-body user devices (e.g., smart glasses (SG) **108**; earphone (EP) **110**; smart watch (SW) **112**; smart phone (SP) **114**); (ii) a user ID linked to the individual devices, allowing user devices having a common user to be identified; (iii) a network connection status indicating if a wireless network communications link is currently active (e.g., connected) or inactive (e.g., not connected) between the sound system **118** and the respective devices; (iv) an indication of whether the associated device has an on-board microphone; and (v) a list of output

sound property settings for each of the associated devices (which in turn correspond to the sound properties that are presented to the device user). In some examples where a common user ID is associated with a set of devices, then the output sound characteristic settings for all of those devices may be set to the same values.

[0062] In some examples, the output sound property settings for the associated devices may be pre-set to default values. In some examples those default values may be device or user specific, based for example on the most recent settings associated with that particular device's ID or User ID. In some examples the default settings may be set to factory settings or user pre-set settings.

[0063] In the illustrated example, the operations **300** can be triggered by the occurrence of a trigger event from a set of one or more defined trigger events. For example, these trigger events could include: the sound system **118** is activated to start playing audio; the sound system output system switches from one audio program to another (e.g., corresponding to a video channel or program streaming change by smart TV **104**); detection of a newly active network connection with an on-body user device; and a user input through a user interface. In some examples, operations **300** can be performed periodically when sound system **118** is generating a sound output as part of its normal operation. For example, during its normal operation, the sound system **118** can be outputting a multi-channel sound sequence (e.g., front right, front center and front left sound channels of a movie sound track) that corresponds to a video output (e.g., a movie) that is being displayed on video display **122** of the smart TV **104**.

[0064] As indicated at Block **302**, as part of first device operations **302**, the smart device interaction module **160** of the sound system **118** identifies the presence of one or more microphone equipped on-body user devices within a listening range of the sound system **118**. In an example embodiment, the smart device interaction module **160** of sound system **118** is configured to continuously maintain associated smart device data **162**, including for example the data shown in above Table 1. Based on such data, the sound system **118** is able to identify what microphone equipped on-body user device(s) it currently has an active local wireless network connection in place with. The presence or absence of a network connection can function as an indication of the presence or absence of one or more associated target on-body devices within a listening range of the sound system **118**. In some examples, the type of network connection may be used to determine device presence; for example, a direct peer-to-peer network communication link **134** (for example Bluetooth™) may be considered to be indicative of the presence of an on-body user device, while an indirect connection view a WiFi base station **120** may not be considered to be indicative of the presence of an on-body user device within a listening range of the sound system **118**.

[0065] In some examples, sound system **118** may determine that there are multiple microphone equipped on-body user devices within a listening range (e.g., smart glasses **108**; earphone **110**; smart watch **112**; and smart phone **114**). In such scenarios, the sound system **118** may be configured to select one such on-body user device as a representative target on-body user device based on a defined hierarchy of such devices as a means for representing a user listening experience. For example, head-mounted on-body user devices such as smart glasses or earphone **110** may be ranked as 1.sup.st choice and 2.sup.nd choice, respectively, due to user ear proximity, with on-body user devices that are further from the user's ears such as smart watch **112** and smart phone **114**, being ranked as 3.sup.rd choice and 4th choice, respectively. In some examples, the user ID associated with the connected on-body user devices can also be used to rank the on-body user devices. For example user **102** may be pre-defined as a higher priority user than a further user that has an on-body user device that is also within a listening range of the sound system **118**. Thus, one or a combination of an on-body user device type and a user ID associated with the on-body user device can be used to determine a relative ranking of the on-body user devices that are determined to be within a listening range of the sound system **118**.

[0066] Sound system **118** selects one or more of the connected microphone equipped user devices as a target on-body user devices. In some examples where ranking is applied, only the 1.sup.st

ranked on-body user device is identified as a target on-body device for further of the operations **300**. In some examples, multiple on-body user devices can be selected as target on-body devices. [0067] The scenario will now be described for the case where a single on-body user device has been identified in Block **302** as the target on-body device, namely smart glasses **108**. As indicated in Block **304**, the sound system **118** is configured to send a wireless message (e.g., a “record request”) for the smart glasses **108**. The record request is sent as an RF signal **605** using the smart home environment wireless network **129** (e.g., using a peer-to-peer communication link **134** or a WiFi communication link **130**, or a combination thereof).

[0068] In some examples, as indicated at Block **306** the target on-body user device (e.g., smart glasses **108** in the illustrated example) is configured to receive the RF signal **305** and decode and process the record request message. In at least some examples, the smart glasses **108** will determine if it is able to perform the record request (e.g., determine that its on-board microphone **116** is available to perform the record request and not currently in use for a different function), and will send an wireless RF signal **307** via one or more communications links of wireless network **129** for sound system **118** that includes an indication that the smart glasses **108** are available (or not available) to perform the record request. In such scenarios, the sound system **118** will monitor for the acknowledgement (ACK) RF signal **307** from the smart glasses **108**, and if the smart glasses **108** are determined to not be available to perform the request, return to Block **302** to select a different target on-body user device.

[0069] As indicated at Block **308**, after sending the record request and determining that the smart glasses **110** are available to process the record request, the sound system **118** causes speakers **124**, **126** and **128** to generate an acoustic wave sound segment **310** for a sound segment duration using a known set of sound properties (for example, using the sound properties specified in Table 1). In some examples, the sound segment **310** that is played may be a segment of an on-going audio output that is being played during a normal operation of the sound system **118** (e.g., the sound for a video program or music program that is currently being played). The sound system **118** can store an electronic version of reference audio signal **309** of the sound segment for future processing purposes.

[0070] In some examples, the sound segment **310** that is played may be a dedicated sound sample generated based on a stored custom reference audio signal **309** that has been created for the purpose of operations **300**. For example, the sound segment **310** could be a sine sweep. A sine sweep is a sine function that gradually changes frequency over time. The sine sweep can also be called “sinusoidal sweep,” “frequency sweep”, or “chirp”, generated using the function $x(t)=\sin(2\pi(f_0*t+((f_1-f_0)/2T)*t\{\text{circumflex over ()}\}2))$. For example, the sound frequency can be changed from f_0 to f_1 over the time T , where $f_0=20$ Hz, $f_1=20000$ Hz, and $T=60$ s. The frequency sweeping sound segment **310** can be played solely or mixed with other sounds, including sounds of a normal operation of the sound system **118**. The sound samples can include channel specific sounds for output by each of the respective speakers **124**, **126** and **128**.

[0071] As indicated at Block **312**, upon receiving and acknowledging the record request message, the smart glasses **108** causes its embedded microphone **116** to be activated for a defined sound segment duration in order capture the sound segment **310**. The microphone **116** converts received acoustic waves corresponding to the sound segment **310** into a user device audio signal **317** that can be recorded and electronically stored in memory. The user device audio signal **317** will include information that indicates one or more properties of the sound segment that has propagated from the sound system **118** to the user **102**. In some examples, the defined sound segment duration may be specified in the record request message; in other examples, it may be a preset value known to smart glasses **108**. In some cases, the target on-body user device (e.g. smart glasses **108**) may include multiple microphones **116**, in which cases multiple microphones may be activated for the sound segment duration for respectively recording the sound segment to generate recorded user device audio signal **317**. It will be noted that in the illustrated example the on-body user device

microphone **116** is only activated for the defined sound segment duration. This may help alleviate privacy concerns of users who do not want to be continually monitored.

[0072] As indicated at Block **314**, the target on-body user device (e.g. smart glasses **108**) then transmits an RF signal **316**, embedded with a copy of the user device audio signal **317**, through the wireless network **129**. As indicated at Block **318**, sound system **118** receives and decodes the RF signal **316** to retrieve and store a local copy of recorded user device audio signal **317**.

[0073] As indicated in Block **320**, the sound system **118** processes the recorded user device audio signal **317** to determine if any action should be taken such as adjusting the sound properties of the sound that is being generated by sound system **118**. As noted above, the user device audio signal **317** will include information that indicates one or more properties of the sound segment that has propagated from the sound system **118** to the user **102**. In some examples, the processing can include comparing parameters of the stored reference audio signal **309** with corresponding parameters of the recorded user device audio signal **317**, thereby enabling properties of the propagated sound segment to be extracted. In some examples, the processing may be based on comparison of parameters of the recorded user device audio signal **317** with defined thresholds.

[0074] By way of example, the sound system **118** can be configured to determine if adjustments to sound volume is required by comparing sound magnitude parameters extracted from the stored reference audio signal **309** with those extracted from the recorded user device audio signal **317**. If the magnitude parameters of the recorded user device audio signal **317** fall below a threshold level relative to the comparable magnitude parameters of stored reference audio signal **309**, a determination may be made that a volume property of the sound generated by the sound system **118** needs to be increased. These magnitude parameters can also be considered on a channel by channel basis based on analysis of channel specific sound outputs of the respective speaker channels to determine if channel balance and fade property adjustments are required.

[0075] In the case of channel delay properties that can be provide an enhanced spatial audio experience, in at least some examples, a physical location of the user **102** relative to the sound system **118** can be deduced by detecting changes in the relative channel timing in the recorded user device audio signal **317** and the stored reference audio signal **309**. For example, the timing of sounds generated by the respective front left, center and right speakers **124**, **126** and **128** may be set based on an assumption that the user **102** is located front and center of the sound system **118**. By comparing the relative time differences between channel specific sounds represented in the recorded user device audio signal **317** relative to those of the stored reference audio signal **309**, actual variations between an intended and actual time-of-flight of the acoustic waves from speakers **124**, **126** and **128** to the recording on-body microphone **116** can be determined. This information, which is indicative of the actual physical location of the user **102** relative to the sound system **118**, can be used to compute adjustments to the channel delay properties of generated sounds in order to enhance the spatial audio experience of the user **102**.

[0076] Regarding frequency band equalization properties, in some examples a frequency response (i.e., a range of frequencies or musical tones) of recorded user device audio signal **317** (which represents an actual frequency response) and the stored reference audio signal **309** (which can represent a desired frequency response) can be compared to determine differences between an actual response and a desired response and suitable equalization property adjustments calculated to minimize the difference.

[0077] As indicated in Block **322**, in an example embodiment the actions determined in Block **320** are performed. For example, sound property adjustments that are computed by the smart device interaction module **160** of the sound system **118** are applied to control the sound controller **150** to effect the adjustments.

[0078] In some examples, the operations **300** may be repeated on an on-going periodic basis to provide continuing dynamic adjustments to the user listening experience. The operations **300** may be repeated with sufficient frequency to provide real-time adjustments that correspond to changes

in the user's relative position relative to the sound system **118** and/or background noises.

[0079] In some examples, in Block **302** multiple on-body user devices may be selected as target on-body user devices. In such cases, the target on-body user device operations **303** may be performed independently at each of the selected target on-body user devices and the sound system **118** will receive respective recorded audio signals from each of the target on-body user devices for processing. Sound property adjustments can be computed based on the collective processing results.

[0080] In some examples the multiple target on-body user devices may be associated with multiple users, with each user having a different relative location to the sound system **118**. Such a scenario may for example occur when several people are in a room using the smart TV for a video call. In multi-user scenarios, the sound system **118** can be configured to selectively optimize one of the sound properties. For example, the sound magnitude parameters of the respective recorded user device audio signals **317** can each be independently analyzed to ensure the sound volume property is sufficiently high to enable all of the users to hear the sounds output by the speakers **124**, **126**, **128** of the sound system **118**. In at least some examples, the above described time-of-flight analysis can be performed to estimate the relative locations of individual users to the respective speakers **124**, **126**, **128**, and the resulting data used to individually adjust the individual speaker volumes so as to provide the collective optimal sound volume for all of the participating users. Further, the inclusion of unique speaker channel sounds in the sound segment can enable the volume property magnitudes to be determined for each of the individual speakers **124**, **126**, **128** for each of the participating target on-body user devices. This data can be used to individually adjust the individual speaker volumes so as to provide the collective optimal sound volume for all of the participating users.

[0081] In some examples, the actions that are determined in Block **320** and performed in Block **322** can be or include actions other than sound property adjustments. For example, in some implementations processing of the recorded user device audio signal **317** may include comparing sound magnitude properties to one or more defined thresholds to determine if an unsafe listening experience (e.g., potentially ear damaging volume is occurring. In the event that an unsafe listening experience is detected, automated actions could include one or more of: displaying a volume adjustment suggestion on display **122**; causing sound output to pause; sending an email or text notification to a defined contact (e.g., a parent) advising of the unsafe listening experience; and adjusting the output volume down to a lower level.

[0082] In some examples, the thresholds to determine if an unsafe listening experience exists could be stored as part of the associated smart device data **162** and could be user and/or user device specific thresholds. For example, a volume based threshold for an on-body user device associated with a child may be preset through a user interface to be lower than that for an on-body user device associated with a hearing impaired senior.

[0083] In some examples, the thresholds for detecting an unsafe listening experience could be based on frequency response properties of the recorded user device audio signal **317**.

[0084] In the illustrated example, the sound system **118** selects the target on-body user device(s) to send a record request to. However, in some examples, an on-body user device may be used to initiate operations. By way of example, FIG. **4** illustrates a set of operations **400** that are the same as operations **300** with the exception of difference noted below. As indicated in Block **406**, in the case of operations **400**, the on-body user device (e.g., smart glasses **108**) sends an RF signal **405** that includes a calibration request for sound system **118** via wireless network **129**. The request may for example be initiated by an input by user **102** to the on body user device. For example, the user may issue a verbal instruction "Smart Glasses, calibrate Smart TV sound system output" that is picked up by the microphone **116** of smart glasses **108** and triggers the request. Upon receiving the request, the sound system **118** commences performing operations **401**. Thereafter, the operations represented by Blocks **308** and onward in FIG. **4** can be identical to those described above in

respect of FIG. 3.

[0085] In some alternative examples, recorded user device audio signal **317** may not be transmitted to the sound system **118**, and some or all of the processing operations associated with Block **320** may instead be performed at the target-on-body user device. In this regard, FIG. 5 illustrates operations **500** corresponding to an example embodiment where an on-body user device (e.g., smart glasses **108**) are configured to periodically (or in response to a user input or other defined trigger event) activate its microphone **116** to capture a sound segment (Block **510**) that has been generated by the sound system **118** during its normal operation (Block **508**). The on-body user device may process the resulting user device audio signal **317** locally (Block **512**) by comparing properties of the user device audio signal **317** to locally stored audio property thresholds to determine if any actions are required to adjust a listening experience of the user associated with the on-body user device. For example, sound magnitude properties of the user device audio signal **317** can be processed to determine if they exceed one or more maximum thresholds in which case a volume reduction action is required, or fall below one or more minimum thresholds, in which case an increase volume action is required. Based on the processing results, the on-body user device transmits an RF signal via network **129** requesting the required action (Block **514**). In the example of FIG. 5, the sound system **118** receives and implements the requested action (Block **522**), thereby enhancing the user's listening experience.

[0086] The above examples have focused on enhancing the user's listening experience. In some implementations the respective smart devices of smart home environment **100** can be configured to enhance a user's speaking or "talk" experience by selectively using audio signals that are generated at multiple device microphones for the same user generated sounds. In this regard, FIG. 6 illustrates a flow diagram showing operations **600** performed by a first smart device (first device operations **601**) and an on-body user device (target on-body user device operations **303**) to enhance a user's talk experience, according to a first example embodiment. For illustrative purposes, operations **600** will be explained in the context of sound system **118** functioning as the first smart device and smart glasses **108** functioning as the target on-body user device. In the illustrated example, sound system **118** includes an associated dedicated local microphone **116** for capturing sound within the local smart home environment. For example, sound system microphone **117** may be connected to an I/O interface **204** of the smart device processor system **200** of the sound system **118** by a wired or wireless communication link. Sound system microphone **117** can, for example, be integrated into a housing of the smart TV **104** (or a housing of the sound system **118**, in the case where the sound system **118** has an independent housing), or may be a stand-alone microphone operatively connected to the sound system **118**.

[0087] The sound system microphone **117** is configured to convert sound into an audio signal input for the sound system **118** for further processing. In some use scenarios, the sound can include a voice command or voice input that is intended to cause the smart device processor system **200** of the sound system **118** to perform an action based on the voice command or voice input. In such cases, the further processing includes converting the audio signal input into a computer actionable input. Examples of such voice commands or inputs can include, for example "Increase Volume"; "Change Channel"; "Record this program"; "Video Call Bob Smith"; "Tell me the weather", and the like. In other use scenarios, the sound can include voice and other audio range sounds that are part of a dialogue in which the smart device processor system **200** of the sound system **118** acts simply as an intermediary that transmits the audio signal input along to a further device. For example, in the context of an on-going video conference call, the sound captured by sound system microphone **117** is converted into an audio signal for transmission with a video signal over one or more networks to a counterpart video conference-call device for consumption by a further party to the call.

[0088] In each of the above cases, the input sound received at sound system microphone **117** is converted into a local audio signal. In example embodiments, the operations **600** can be triggered

by one or more defined trigger events. In an illustrative example, the smart device processor system **200** of the sound system **118** is configured to monitor one or more sound properties of the local audio signal output by sound system microphone **117** and determine when one of more of those properties fails to meet a defined performance criteria. For example, determining if a performance criteria is met can be based on comparing a volume parameter extracted from the local audio signal to a defined threshold. Failure to meet the performance criteria can be indicative that the sound system microphone **117** is not sufficiently capturing the sound input being provided by a user **102**. In an illustrative example, the failure of the sound properties of the local audio signal output to meet a defined performance criteria is a trigger event than causes the sound system **118** to commence operations **600**. A purpose of operations **600** is to enable the sound system **118** to supplement the sound input it is receiving at its local microphone **117** with sound inputs captured by the respective microphones **116** of one or more of the on-body devices that are present in the smart home environment **100**.

[0089] As indicated at Block **602**, when a trigger event occurs, the smart device interaction module **160** of the sound system **118** is configured to identify and select one or more microphone equipped target on-body user devices within a speaking range of the sound system **118**. This can be done in a similar manner as described above in respect of Block **302** to select a microphone equipped target on-body user device that is within a listening range.

[0090] The scenario will now be described for the case where a single on-body user device has been identified in Block **602** as the target on-body device, namely smart glasses **108**. As indicated in Block **604**, the sound system **118** is configured to send a wireless message (e.g., a “record request”) for the smart glasses **108**. The record request is sent as a wireless electromagnetic (EM) radio frequency (RF) signal **305** using the smart home environment wireless network **129**. For example, RF signal **305** could be sent using a peer-to-peer communication link **134** or a WiFi communication link **130**, or a combination thereof (e.g., base station **120** can relay the record request message from sound system **118** to smart phone **114** which in turn uses a peer-to-peer link to relay the message to smart glasses **108**).

[0091] In some examples, as indicated at Block **606** the target on-body user device (e.g., smart glasses **108** in the illustrated example) is configured to receive the RF signal **605** and decode and process the record request message. In at least some examples, the smart glasses **108** will determine if it is able to perform the record request (e.g., determine that its on-board microphone **116** is available to perform the record request and not currently in use for a different function), and will send an wireless RF signal **607** via one or more communications links of wireless network **129** for sound system **118** that includes an indication that the smart glasses **108** are available (or not available) to perform the record request. In such scenarios, the sound system **118** will monitor for the acknowledgement (ACK) RF signal **607** from the smart glasses **108**, and if the smart glasses **108** are determined to not be available to perform the request, return to Block **602** to select a different target on-body user device.

[0092] As indicated at Block **608**, the sound system **118** proceeds with using its local microphone **117** to capture sound input and generate a respective recorded local audio signal **609**. In an example scenario, the sound input that is captured corresponds to a sound segment spoken by the user **102** that is wearing the smart glasses **108**. Thus, the recorded local audio signal **609** will include properties of the sound segment as propagated from the user's mouth to the local microphone **117**.

[0093] As indicated at Block **612**, upon receiving and acknowledging the record request message, the smart glasses **108** causes its embedded microphone **116** to be activated to also capture the sound input. The microphone **116** converts received acoustic waves corresponding to the sound segment into a user device audio signal **617** that can be recorded and electronically stored in memory. In some cases, the target on-body user device (e.g. smart glasses **108**) may include multiple microphones **116**, in which cases multiple microphones may be activated for the sound segment duration for respectively recording the sound to generate recorded user device audio signal

617.

[0094] As indicated at Block **614**, the target on-body user device (e.g. smart glasses **108**) then transmits an RF signal **616**, embedded with a representation of the user device audio signal **617**, through the wireless network **129**. As indicated at Block **618**, sound system **118** receives and decodes the RF signal **616** to retrieve and store a local version of recorded user device audio signal **617**.

[0095] As indicated in Block **620**, the sound system **118** processes the user device audio signal **617** and the corresponding local audio signal **609** to extract information for generating a final audio signal **621**. Known signal processing techniques for merging independently captured audio signals corresponding to the same input sound can be applied to provide a final audio signal **621** that can have enhanced qualities relative to one or both of the user device audio signal **617** and the corresponding local audio signal **609**. The final audio signal **621** can then be used for the purpose intended for the local audio signal **609** (i.e. as a voice command or input, or for transmitting a representation of a dialog).

[0096] Accordingly, operations **600** can enhance the talking experience of user **102** as the sound system **118** works dynamically and automatically with the target-on-body user device (e.g., smart glasses **108**) to compensate for a poor quality sound input at the microphone **117** of the sound system **118**. In at least some use scenarios, an enhanced final audio signal **621** can be generated by the sound system **118** such that the user **102** is not required to change their speaking volume or to change their location to move closer to the microphone **117** of the sound system **118**.

[0097] In some examples, multiple on-body user devices may be selected as target on-body user devices. In such cases, the target on-body user device operations **603** may be performed independently at each of the selected target on-body user devices and the sound system **118** will receive respective recorded user device audio signals **617** from each of the target on-body user devices for processing. The resulting final audio signal **621** can be a combination of audio data extracted from one or more of the user device audio signals **617** and the local audio signal **609**. Such a system can be particularly useful when multiple users are using a common smart device (e.g., Smart TV sound system **118**) for a joint video conference call.

[0098] Although certain operations are described above as being performed by the smart device processor systems **200** of specific smart devices, in alternative embodiments various operations may be performed or distributed partially or completely on other smart devices within the smart home environment **200**.

[0099] It will be appreciated that the disclosed smart home environment **100** can, in some scenarios, provide users with optimized experience without users' manual operation of individual smart devices that are present in the environment. In overview, the present disclosure is directed resolving the existing problem that that in a smart home environment it can be difficult for participating smart devices to accurately determine what sounds a user is hearing or making. The disclosed embodiments provide a solution by using the microphone of an on-body user device to enhance the user experience. Given that these on-body microphones will typically be close to the user's ear and mouth, the recordings from these microphones can be used to capture the sound the user is hearing or making. These recorded audio signals can be used as the input for other smart devices to provide the user with intelligent services. In the case where the user is hearing the sound, the properties of the sound as propagated from the originating smart device to the user will be embedded in the recorded audio signal. In the case where the user is speaking the sound, the properties of the sound as propagated from the user to a local microphone of the smart device will be embedded in the audio signal that is recorded at the smart device, whereas the audio signal as recorded at the microphone of the on-body user device will embed the properties of the pre-propagated sound at its origin (e.g., the user's mouth).

[0100] In each of the listen and talk scenarios, one or more properties of a sound segment propagated between a smart device and a user can be determined based on an audio signal

generated by a microphone of an on-body user device of the user. In the case of the listen scenario, the sound propagated between the smart device and the user is the sound that is generated by the output speakers of the smart device and captured by the microphone **116** of the on-body user device. In the case of the talk scenario, the sound propagated between the smart device and the user is in the opposite direction and is the sound that is generated by speaker's mouth and captured by the local microphone **117** of the smart device. In both cases, properties of the propagated sound can be extracted based on the audio signal generated by the microphone of the on-body user device. An action can be performed based on determining the one or more properties of the propagated sound. [0101] Although the present disclosure describes methods and processes with steps in a certain order, one or more steps of the methods and processes may be omitted or altered as appropriate. One or more steps may take place in an order other than that in which they are described, as appropriate.

[0102] Although the present disclosure is described, at least in part, in terms of methods, a person of ordinary skill in the art will understand that the present disclosure is also directed to the various components for performing at least some of the aspects and features of the described methods, be it by way of hardware components, software or any combination of the two. Accordingly, the technical solution of the present disclosure may be embodied in the form of a software product. A suitable software product may be stored in a pre-recorded storage device or other similar non-volatile or non-transitory computer readable medium, including DVDs, CD-ROMs, USB flash disk, a removable hard disk, or other storage media, for example. The software product includes instructions tangibly stored thereon that enable a processing device (e.g., a personal computer, a server, or a network device) to execute examples of the methods disclosed herein.

[0103] The present disclosure may be embodied in other specific forms without departing from the subject matter of the claims. The described example embodiments are to be considered in all respects as being only illustrative and not restrictive. Selected features from one or more of the above-described embodiments may be combined to create alternative embodiments not explicitly described, features suitable for such combinations being understood within the scope of this disclosure.

[0104] All values and sub-ranges within disclosed ranges are also disclosed. Also, although the systems, devices and processes disclosed and shown herein may comprise a specific number of elements/components, the systems, devices and assemblies could be modified to include additional or fewer of such elements/components. For example, although any of the elements/components disclosed may be referenced as being singular, the embodiments disclosed herein could be modified to include a plurality of such elements/components. The subject matter described herein intends to cover and embrace all suitable changes in technology.

[0105] As used herein, statements that a second item is “based on” a first item can mean that characteristics of the second item are affected or determined at least in part by characteristics of the first item. The first item can be considered an input to an operation or calculation, or a series of operations or calculations that produces the second item as an output that is not independent from the first item.

Claims

1. An automated method comprising: processing a user device audio signal representing a sound segment received at a microphone of an on-body user device of a user, wherein the sound segment is propagated as an acoustic wave between a smart device and the user; and performing an action based on the processing.
2. The method of claim 1 wherein the sound segment is generated by one or more speakers of the smart device and propagates from the one or more speakers to the user.
3. The method of claim 2 wherein processing the user device audio signal comprises extracting

information about one or more sound properties of the sound segment from the user device audio signal and performing the action comprises causing an adjustment of the one or more sound properties for future sounds generated by the one or more speakers.

4. The method of claim 3 wherein the one or more sound properties includes a volume.

5. The method of claim 3 wherein the one or more sound properties includes a frequency equalization.

6. The method of claim 3 wherein the one or more sound properties includes a relative balancing of multiple sound channels.

7. The method of claim 3 wherein the one or more sound properties includes a relative time delay between multiple sound channels.

8. The method of claim 3 wherein extracting information about one or more sound properties of the sound segment comprises comparing a reference audio signal used to generate the sound segment with the user device audio signal.

9. The method of claim 8 comprising: receiving by the smart device, through a wireless network, a transmitted copy of the user device audio signal, wherein processing the user device audio signal and performing the action are each performed by a processor system of the smart device.

10. The method of claim 1 comprising: sending by the smart device, through the wireless network, a message for the on-body user device requesting the on-body user device to record the sound segment to generate the user device audio signal.

11. The method of claim 2 comprising generating, by the on-body user device, the user device audio signal, wherein processing the user device audio signal and performing the action are each performed by a processor system of the on-body user device and performing the action comprises causing an action request message to be sent through a wireless network communication link for the smart device.

12. The method of claim 1 wherein the sound segment comprises a frequency sweep.

13. The method of claim 1 wherein processing the user device audio signal comprises determining based on one or more parameters extracted from the user device audio signal when an unsafe listening experience exists for the user, and performing the action comprises, upon determining the unsafe listening experience exists, performing one or more actions to end the unsafe listening experience, including one or more of: causing a reduction in sound volume; adjusting a sound frequency equalization; causing sound to stop being generated; causing a message about the unsafe listening experience to be displayed on a display screen; and causing a message about the unsafe listening experience to be transmitted to a further device.

14. The method of claim 3 comprising storing user device specific settings for the on-body user device and a plurality of further user devices, wherein performing the action is further based on the user device specific settings stored in respect of the on-body user device.

15. The method of claim 1 wherein performing the action comprises causing one or both of a listening experience and a talking experience of the user to be automatically adjusted without requiring user action to effect the adjustment.

16. The method of claim 1 wherein the sound segment originates from a mouth of the user and propagates from the user to a microphone of the smart device, the method comprising: generating, by the smart device, a local smart device audio signal representing the sound segment as received at the microphone of the smart device; receiving by the smart device, through a wireless network, a transmitted copy of the user device audio signal; processing the local smart device audio signal in combination with processing the user device audio signal; wherein the action comprises generating a final audio signal based on the combined processing of the local smart device audio signal and the user device audio signal.

17. The method of any claim 1 wherein the smart device comprises one of a smart TV, an interactive smart speaker, a smart appliance, a smart sound system, and the on-body user device comprises one of a smart watch, a fitness tracker, smart glasses, a headset, an earphone, smart

clothes and a hand-held mobile device.

18. A processor system comprising: a processor; and a memory storing machine-executable instructions thereon which, when executed by the processor, cause the processor system to: process a user device audio signal representing a sound segment received at a microphone of an on-body user device of a user, wherein the sound segment is propagated as an acoustic wave between a smart device and the user; and perform an action based on the processing.

19. A non-transitory processor-readable medium having machine-executable instructions stored thereon which, when executed by a processor, cause the processor to: Process a user device audio signal representing a sound segment received at a microphone of an on-body user device of a user, wherein the sound segment is propagated as an acoustic wave between a smart device and the user; and perform an action based on the processing.
